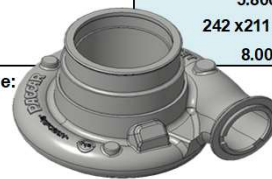


RFQ No.	2425-328	Date	7-Feb-25
Customer	CTT	Annual volume (Nos)	33,000
Final Part No.	6511292	Annual volume including rejection (Nos)	37,950
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)	2.960
Description	Meredian Housing Compressor	Casting Weight (Kg)	3.800
Alloy	E4-01-240 (C355-T71)	LBH (mm)	242 x211 x134
Alternate alloy proposed by SCL	-	Takt time (min)	8.00



Add info of R/C:-				Part image:		
Output/ m/c details:				No of cavities/ Loading	Cycle Time (min)	Output /hr
Sand core 1				2	5.0	24
Sand core 2						0
Sand core 3						0
Sand core 4						0
Decore				1	2.0	30
Casting				2	6.0	20
Cutting 1				1	2.0	30
Cutting 2						0
Linishing				1	2.0	30
Shot blasting						192

	
Casting Category	Medium-Hor
No of casting cells planned	1
Weight of Sand core/ part (kg) (f any)	2.75
Man power / shift/ cell	6
Floor space /cell (Sq.m) (includes Die casting, Post casting, WIP, Sand core, finished goods, HT, SH/blast, Melting)	676
Type of shot blasting (Hanger type /AIO2/-Tumbling/-Vibro-finishing)	Hanger Type
Surface coating involved (if any)	-

Capital Investments (Rs. Lac)		Utilisation%	No of M/c/s/ No of Units	Amount /cell	Total Cost
Sand core	Core shooting M/c -1	32%	1	45.0	45.0
	Core shooting M/c -2				0.0
	Core Painting Oven (if any)		0		0.0
	Thermal Decoring M/c				0.0
	Vibro Decoring Machine	25%	1	20.0	20.0
Casting	Tilting Machine		1		0.0
	Vertical Machine	38%		25.0	25.0
	Stand type Machine				0.0
	Hydraulic Power Pack			5.0	5.0
	Holding furnace		2	14.0	28.0
Post casting	Band saw machine				0.0
	Circular saw cutting machine	25%	1	60.0	60.0
	Riser cutting machine				0.0
	Knock out Press		1	3.5	3.5
	Linishing Machine with Dust Extractor	25%	1	2.2	2.2
	Shift code punching Machine		1	3.0	3.0
	Welding Machine		0	0.0	0.0
	Bend Removal Press				0.0
Automation	Auto Laddle		1	0.0	0.0
	Robot Pouring & Extraction		1	60.0	60.0
	Robot cell (Pouring robot, Quenching, Cutting m/c, cell forming)		0	0.0	0.0
	Component Extractor or Catcher		1	0.0	0.0
Others	Endoscope		1	8.0	8.0
	Special Core handling / Testing		1	12.0	12.0
	Auto Air cleaning				0.0
Common Facilities	Shot blasting M/c	4%		Refer Note	0.0
	Melting furnace	1%	1	80.6	80.6
	Melting furnace accessories (Fork lift, ATL, Degassing unit, Dross trolley, scrubber unit)			37.0	37.0
	Heat Treatment Furnace (incl crane)				Refer Note
	Heat Treatment BatchCode Punching M/c		1	6.0	6.0
	Core placement fixture		1	2.0	2.0
	Total Investment				397.3
Die	Die and corebox (if any) No of Dies & cost (Rs. Lac)		1		
	Die life (shots)				50,000
	Core box life (shots)				50,000
Testing Cost Details: (Rs.)					
Chemical Testing cost/ part				4.0	
X Ray Testing cost / part					
Tensile Testing cost / part				5.8	
Microstructure Testing cost / part				6.5	
Hardness Testing cost / part				0.1	
3D Inspection cost / part				0.8	
Porosity Testing cost / part					
Others (if any)					
Total Testing Cost/ Part				17.2	

Operating Cost (Rs. Lac)		Amount (Rs.)
Sand core	Dressing Station	0.4
	Core Painting Station	0.8
	Core Handling Stand	1.2
	Core Cavity Handling Trolleys	2.4
	Core visual inspection / Table / tools	0.8
	Vibro Decoring Hammer	1.0
Casting	Vibro Decoring Fixture	0.8
	Casting Table	0.3
	Casting & Sand collection Trolley	0.3
	Furnace Dross, Rejection trolley & Ladle set	0.5
	Cutting Fixture (Circular/ Riser/ Band)	0.8
Post Casting	Riser Collection Trolley	1.5
	Fettling table with Indexing Head	0.8
	Indexing Fixture	0.3
	Fettling Table	0.3
	Fettling Booth	0.5
	Cell Formation Cost	-
	Filing & Grinding Tools	0.5
	Final Inspection Table (Post cast+ HT)	0.3
	Storage Pallets	0.2
	Gauges	0.5
	DM Water for Dycoat	0.0
	Shift code punching Fixture	0.7
	Welding Booth	0.0
	Welding Fixture	0.0
	Fan for Operators (Areas at Core, Decore, Casting, Fettling, HT (if any), Shblast (if any))	0.1
	Material handling (Crates+ Trolley)	
	Material handling (Trolley - mould type)	0.0
	Bend Removal Tool/ Fixture/ Gauge	
	Profile Filter	3.0
	Common Facilities	Shot blasting Hanger
Stainless Steel shots for the volume		
HT Batch Code Punching Fixture		1.0
HT Basket cost		1.5
Die	Jet Cooling cost (Die Temp. Contr)	
	Die service consumable	0.0
Total Investment		21.0
Power Rating Details: (Kw/hr)		
Casting Cell (Incl Core shoot, Decore, Holding furnace, Casting, Power pack, Cutting, Linishing, Robot, Extractor)		219.0
Melting Furn. Capacity	1000 T	24.6
Heat treatment		Refer Note
Shot blasting		Refer Note
Discussed with: Mohan / Sathish kumar.		
Similar part/ Estn referred:		

CostEstimation

Basic Details

Business Feasibility

Technical Feasibility

Die Design

Die

Cost Estimation

Purchase

Pricing

Quote

Customer Feedback

Others

RFQ No.:	2425-328	Creation Date :	11/01/2025	Customer Group :	Cummins Exports	Annual Volume :	33000
Part Detail :	HOUSING,COMPRESSOR (6511292)			Project Name :	Meridian	Weight :	3.8
Manufact Location :	SCL India			Die Design User :	Mohan S		

Technical

Die design

Part image upload

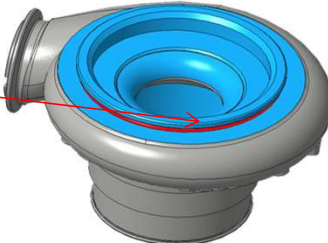
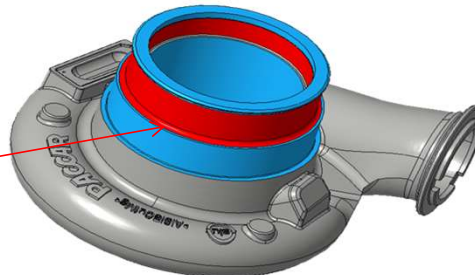
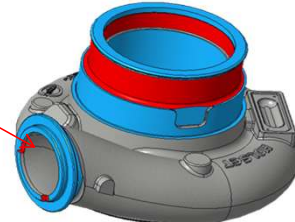
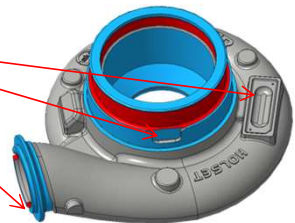
☐ Reject RFQ

Design Feasibility

Quality Factors	Status	Remarks	Remarks (sec head)	Action
No of cavities in Die (Family mould/ Part mould)	Yes	2		
No of metal cores wrt. die face	No			
Casting Cycle Time(In sec)	Yes	360 Sec (180 Sec per cavity)		
Sand core Cycle Time(in sec)	Yes	300 Sec (150 Sec per cavity)		
Decorng Cycle Time(in sec)	Yes	120 Sec		
Cutting Cycle Time(in sec)	Yes	120 Sec		
Linishing Cycle Time(in sec)	Yes	90		
No of cavities in core box & \ (Same part) / B (Family mould)	Yes	2\		
No. of sand cores required	Yes	1		
Parting type Hor/ Ver)	Yes	Ver		
No of core box required	Yes	1		
Weight of sand core (kg)	Yes	2.5 Kg		
Machine type:	Yes	Vertical-VDCM		
Pouring type -	Yes	Vertical		
Component extractor-	Yes	Auto		
Cutting m/c type	Yes			
Parting type Hor/ Ver)	Yes			
Core box life	Yes			
Core Painting Oven Requirement	Yes			
Thermal Decorng requirement	Yes			
Welding rework reqt	No			
Shift code marking reqt	Yes			
Shot blasting reqt	Yes			
Anticipation of bend	No			
Gauge cost	No			
Die life	Yes	50000		
Component extractor-	Yes	Auto		
Coating type req. koren /Alumina	Yes	Riotech		
Coating req. grams / part	Yes	10 gms		
No of cavities baffle	No			
Decore Machine type (shaker/Vibro)	Yes	Vibro		
Core shooting Machine type (1824VP/4040VP/3436/VP)	Yes	1824VP		
Baffle cycle time	No			
Tilting Machine spec - (990x520 / 1370x900 / 1600x965)	No			
Vertical Machine spec - (1000x535 / 1205x740 / 1230x760)	Yes	1230 x 760		
Die block size	Yes	800 x 700 x 200		
Endoscope reqt	Yes			
LBH of the part	Yes	241 x 211 x 134		
Total surface area cm2	Yes	228.8		
Yield considered %	Yes	70		
Yield considered %	Yes	70		
Is raiser pad machining required ?	Yes			
If yes, how many raised pad?	Yes	1		
-	Yes			
Masked machining area cm2	Yes			
DieDesign-PDC	No			

Remarks		
User / Approver	Approved Date	Remarks
Mohan S	27/01/2025 11:14:12.147	Twin cavity die & core box considered. Surface roughness 8Ra in volute mention in the drawing. hence AFM process is required. Get the details from AFM team. Wall thickness tolerance provided +1.15mm / -0.37mm
Sathis Kumar S	28/01/2025 09:51:12.047	At the moment we do not have proven process for <8 Ra for HCC. Trials are on going. We can commit upto 13 Ra with special AFS sand. to be discussed with customer.

 SCL-PED		RFQ Estimation for Machining																																														
RFQ No.	2425-328	Date				7-Feb-25																																										
Customer	CTT	Annual volume (Nos)				33,000																																										
Final Part No.	6511292	Annual volume incl. rejection (Nos)				36,300																																										
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)				2.960																																										
Description	Meredian Housing Compressor	LBH (mm)				242 x211 x134																																										
Alloy	E4-01-240 (C355-T71)	Takt Time (min)				8.42																																										
Casting Category	Medium-Hor																																															
Opn. No.	Description	Cycle Time (min)	No of m/cs / cell	Machine Cost (Rs.)	No. of Cells	Amount (Rs.)																																										
20	TURNING CENTER- Diffuser & Inlet M/cng (LT-20)	7.0	1	40,00,000	1	40,00,000																																										
30	TURNING CENTER- Diffuser & Inlet M/cng (LT-20)	5.9	1	40,00,000	1	40,00,000																																										
40	TURNING CENTER- Outlet M/cng- SMALL, MEDIUM (LT-20)	5.0	1	40,00,000	1	40,00,000																																										
50	VMC 450 & 5th AXIS - WTH ROTARY TABLE, XYZ- 600x450x500, TS-900 X 450	7.0	1	89,00,000	1	89,00,000																																										
60	WASHING MACHINE	2.0	1	36,00,000	1	36,00,000																																										
70	DRY CUM WET LEAK TEST	2.0	1	18,00,000	1	18,00,000																																										
80	LASER MARKING WITH 2D SCANNER	2.0	1	24,00,000	1	24,00,000																																										
90	FINAL INSPECTION AREA	2.0	1	2,00,000	1	2,00,000																																										
100	ENDOSCOPE STATION	2.0	1	8,00,000	1	8,00,000																																										
110						0																																										
#REF!						0																																										
	MATERIAL HANDLING IN MACHINE SHOP					0																																										
		Cell Cycle Time:	7.00	Total Capital Expenditure		2,97,00,000																																										
This estimation is valid for 90 days only				F/PED-EST/06 Rev.No. 01 Rev. Dt. 29.10.2014																																												
Operating cost for the given Volume: (Cost in Rs.) <table border="1"> <tr> <td>Cost of Cutting tools</td> <td>3,63,000</td> </tr> <tr> <td>Cost of Tool Holders</td> <td>22,00,000</td> </tr> <tr> <td>Cost of Probing Unit (if any)</td> <td></td> </tr> <tr> <td>Cost of Fixtures</td> <td>23,00,000</td> </tr> <tr> <td>Cost of Gauges</td> <td>6,80,000</td> </tr> <tr> <td>Cost of Material Handling</td> <td>3,00,000</td> </tr> <tr> <td>Coolant oil cost</td> <td>41,382</td> </tr> <tr> <td>DM water cost for coolant, washing & Leak</td> <td>21,780</td> </tr> <tr> <td>Barcode label recurring cost</td> <td></td> </tr> <tr> <td>Impregnation Basket cost</td> <td>0</td> </tr> <tr> <td>Total Oerating Cost</td> <td>59,06,162</td> </tr> </table> <table border="1"> <tr> <td>Cell Cycle Time (Min.)</td> <td>7.00</td> </tr> <tr> <td>Cell Capacity (Nos.)</td> <td>43,682</td> </tr> <tr> <td>Cell Utilisation</td> <td>83%</td> </tr> <tr> <td>No. of Cells</td> <td>1</td> </tr> </table> <table border="1"> <tr> <td>Power rating for cell (kw/hr)</td> <td>115.0</td> </tr> <tr> <td>Man Power / Shift / Cell</td> <td>2</td> </tr> <tr> <td>Floor area required / cell(Sq.M)</td> <td>226.0</td> </tr> <tr> <td>IMP Salvaging %</td> <td>0%</td> </tr> <tr> <td>Setup changeover considered (Y/ N)</td> <td>N</td> </tr> <tr> <td>No of variants planned / cell</td> <td>1</td> </tr> </table>							Cost of Cutting tools	3,63,000	Cost of Tool Holders	22,00,000	Cost of Probing Unit (if any)		Cost of Fixtures	23,00,000	Cost of Gauges	6,80,000	Cost of Material Handling	3,00,000	Coolant oil cost	41,382	DM water cost for coolant, washing & Leak	21,780	Barcode label recurring cost		Impregnation Basket cost	0	Total Oerating Cost	59,06,162	Cell Cycle Time (Min.)	7.00	Cell Capacity (Nos.)	43,682	Cell Utilisation	83%	No. of Cells	1	Power rating for cell (kw/hr)	115.0	Man Power / Shift / Cell	2	Floor area required / cell(Sq.M)	226.0	IMP Salvaging %	0%	Setup changeover considered (Y/ N)	N	No of variants planned / cell	1
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Assumptions/ Notes: 1) Refer additional sheet for remarks pertaining to Machining (2) Estimated based on 3 shift basis only (3) Cost of Stores & Tools, R&M expenses, Installation & Commissioning, Power cost, load factor are captured in pricing by Finance (4) Deatils pertaining to Impregnation process (Cost, Man power, Power, Consumables like sealant etc) considered in pricing based on per kg cost. (5) The cost of Surface treatment (if any) to be obtained from Purchase only, NOT included in this estimation (6) Setup changeover time (if any) considered included in cycle time																																																
EA / KVG					JR																																											
Prepared by					Approved by																																											
This estimation is valid for 90 days only				F/PED-EST/07 Rev.No. 01 Rev. Dt. 29.10.2014																																												

 SCL-PED RFQ Process Planning Sheet			
RFQ No.	2425-328	Date	45695
Customer	CTT	Annual volume (Nos)	33,000
Final Part No.	6511292	Annual volume incl. rejection (Nos)	36,300
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)	2.960
Description	Meredian Housing Compressor	LBH (mm)	242 x211 x134
Alloy	E4-01-240 (C355-T71)	Takt Time (min)	8.42
Input component  Machining process sequence: Machining process sequence: F <u>MACHINE 1 : LATHE</u> SETUP-1 Id forming profile forming  OUTLETTURNING <u>MACHINE 3 : LATHE</u> INLET OD FORMING  <u>MACHINE 3 : LATHE</u> OUTLET OD FORMING  <u>MACHINE 4 : LATHE</u> Drilling Ø6.2 - 1 Nos Drilling Ø1.65 - 1 Nos Neme plate drilling profile form milling Slot milling 			

TVS SCL-PED		RFQ ESTIMATION FOR ASSEMBLY	
RFQ No.	2425-328	Date	07-Feb-2025
Customer	CTT	Annual volume (Nos)	33000
Final Part No.	6511292	Annual volume incl. rejection (Nos)	36,300
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)	2.96
Description	Meredian Housing Compressor	LBH (mm)	3.8
Alloy	E4-01-240 (C355-T71)	Takt Time (min)	8.424
Cycle Time Details:		Assembly	After Assembly Machining
Cycle Time (min)		0.0	0.0
Output Per Hour (Nos)		0	0
Cell Capacity (Nos)		0	0
Cell Utilisation		0%	0.0%
No. of cells required		0	0
BEFORE AFM MACHINING ASSEMBLY		Capex	Operating
Assembly station / Fixture			
Pokayoke system			
Nut runner			
Special Purpose Machine/ Fixture			
Rivetting Machine / Fixture			
Gauges			
Other Accessories (if any)			
Total Assembly Investment		0.00	0.00
Total Assembly Investment for all cells		0.00	0.00
Sealant consumption/ part (if any) (ml) loc tite 648			0.0
Power rating (Kw/hr)			0.0
Is it feasible to use M/cng operator for assembly (Yes / No)			NO
Manpower / shift/ cell			1.0
Floor space required for assembly station (Sq.m/ cell)			0.0
Is after assembly machining involved? (Yes / No)			0.0
BEFORE AFM MACHINING	Capex	Operating	Cycle Time (min)
Vertical M/cng Center			
Turning Centers			
Special Purpose Machine			
Turning Centers			
Washing Machine			
Leak Testing Machine			
Deburring Unit / Air cleaning / Final Inspection			
Tool Holders cost			
Consumable tools cost			
Fixture Cost			
Gauge cost			
Total after assembly machining Investment	0	0	
Total after assembly machining Investment for all cells	0	0	
TOTAL ASSY INVESTMENTS	0	0	
Power rating (Kw/hr)		-	
Is it feasible to use M/cng cell operator for after assembly m/cng (Yes / No)		No	
Manpower / shift/ cell		1.0	
Floor space required for after assembly m/cng cell (Sq.m/ cell)		-	
Machined Part No/ Rev		Child Part Nos:	
Casting Part No/ Rev			
Classification: Confidential Prepared by: FA / KVG This estimation is valid for 90 days only JR Approved by F/PED-EST/05 Rev. No. 01 Rev. Dt. 29.10.2014			

SCL-PED		RFQ REMARKS	
RFQ No.	2425-328	Date	07-Feb-2025
Customer	CTT	Annual volume (Nos)	33,000
Final Part No.	6511292	Annual volume including rejection (Nos)	37,950
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)	2.96
Description	Meredian Housing Compressor	Casting Weight (Kg)	3.800
Alloy	E4-01-240 (C355-T71)	LBH (mm)	242 x211 x134
Alternate alloy proposed by SCL		-	
Casting Remarks: Product design changes required. Casting weight given in the drawing, Machining weight extracted from 3d. Average wall thickness 5.0 mm. Heat treatment basket in -house 50 Nos As per the design input, we have determined that the AFM process is not required for this part. Instead, we will incorporate the following factors into our pricing: 1. Special sand core cost : Rs. 110/kg 2. New hydraulic core shooting process 3. Dedicated trolley for sand core movement. 4. Special care and handling for core testing Please refer to the email below for design reference.			
RE: Reg- AFM Cost required -242-333  Giridhara Gopal (R&D/PADI/SCL) To: Ashok E (NPD/PADI/SCL) Cc: Venugopalan K (NPD/PADI/SCL); Sathish Kumar S (R&D/PADI/SCL); Mohan S (R&D/PADI/SCL); Karthikeyan K (R&D/PADI/SCL) You forwarded this message on 08-02-2025 10:39. Reply Reply All Forward Print More Fri 07-02-2025 17:06 Dear Sir, Customer requires to meet the surface finish of <8Ra without AFM processing. For achieving <8Ra in as cast condition we are conducting trials and need to finalise on the same. For RFQ purpose, the following requirement can be considered: 1. Special sand to be used (₹110/Kg) 2. New Hydraulic core shooting machine to be considered 3. Dedicated trolley for sand core movement to be considered 4. Special care to be taken in core handling and testing Thanks & Regards, B. Giridhara Gopal From: Ashok E (NPD/PADI/SCL) <e.ashok@sundaramclayton.com> Sent: Tuesday, February 4, 2025 4:06 PM To: Giridhara Gopal (R&D/PADI/SCL) <giridhara@sundaramclayton.com> Cc: Venugopalan K (NPD/PADI/SCL) <kvk@sundaramclayton.com> Subject: Reg- AFM Cost required -242-333 Dear Mr. Giridhara Gopal, Request to share the AFM cost for the subjected part, this RFQ done by Mr. Mohan.			
Machining Remarks: Leak testing pressure 2.4 bar @ 1.25 cc/min Required product design change in the machining feasibility clamping and work support will be discussed during RTS. Estimation to be reviewed after RTS / DQR and Impact if any interms of cycle time or cost will be communicated accordingly with reference to engineering changes based on DFM / DFA agreed with customer.			
Assembly Remarks:			

SCL-PED		RFQ ESTIMATION FOR PACKING 334	
RFQ No.	2425-328	Date	7-Feb-25
Customer	CTT	Annual volume (Nos)	33,000
Final Part No.	6511292	Annual volume incl. rejection (Nos)	37,950
Final Part Rev No.	No 2d drawing only 3d	Machined part wt (kg)	2.960
Description	Meredian Housing Compressor	LBH (mm)	242 x211 x134
Alloy	E4-01-240 (C355-T71)		
Expendable - Packing Box qunaity working			
			
Description	Length	Breadth	Height
Part size	242	211	134
Packing box size	1100	910	1000
Allowances	10	10	10
Arrangement 1	4.00	4.00	7.00
No. of components	112		
Arrangement 2	5.00	6.00	4.00
No. of components	120		
Arrangement 3	8.00	3.00	4.00
No. of components	96		
Arrangement 4	4.00	6.00	4.00
No. of components	96		
Arrangement 5	4.00	8.00	4.00
No. of components	128		
Arrangement 6	7.00	3.00	5.00
No. of components	105		
No. of components per box	120		
Protection cap to be considered in pricing.			